

Connor Spinelli

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EDUCATION

B.S. Biomedical Engineering, Honors · Minor in Biomechanical Engineering · Expected May 2027

4+1 M.S. Robotics, University of Delaware · Expected May 2028

Undergraduate GPA: 3.62 – Dean's List (F'23, S'24, F'24, S'26)

Relevant Coursework: Biomechatronics; Intro to Medical Imaging Systems; Biostatistics; Clinical Biomechanics; Medical Device Design; Biological Transport Phenomena; Biomedical Instrumentation; Bioengineering Mechanics

WORK EXPERIENCE

AI Automation Intern | Proscia | *Philadelphia, PA*

June 2026 - Present

- Built a two-stage AI agent pipeline (Python; 13 scripts, 282 tests, 37 design controls) turning Jira release data into user-guide and release-note drafts; verification preserved, 100% claims cited
- Packaged 6 reusable agent skills and 2 MCP connectors (Paligo, Jira, Confluence) behind a non-engineer installer and operator guide; launched an internal AI hub of 17 pages audited weekly by a scheduled agent
- Led workflow discovery sessions with the Product Team to map tasks on a value-vs-effort matrix, identifying high-priority automation candidates based on volume, frequency, and deterministic logic.

Machine Shop Teaching Assistant | Design Studio, Spencer Lab, UD | *Newark, DE*

Aug 2025 - Present

- Guide 100+ students per semester from CAD to finished part on CNC mills, lathes, laser cutters, and 3D printers, catching design-for-manufacture problems before material is cut
- Support sponsored senior design builds for NASA, Merck, and Bloom Energy; own shop safety and compliance for the Design Studio

Undergraduate Researcher | Human Robotics Lab, UD (Delaware INBRE) | *Newark, DE*

June - August 2025

- Designed and fabricated an adjustable, MRI-compatible support structure for a wrist rehabilitation robot in SolidWorks, selecting non-ferrous materials and verifying fit against scanner-bore constraints
- Built a MATLAB script to compare eight EMG pre-processing pipelines on stretch-reflex data and quantified their effect on statistical significance using mixed-model analysis and Tukey HSD

PROJECTS ([Portfolio](#))

Wrist Support Structure for Catheter Insertion - Junior Design

Spring 2026

- Reimagined the current design for a device that can be mass-manufactured and used by medical professionals in the ICU, including wrist extension angle adjustability and an attachment for catheter securement

Instrumented Lamination Probe - Capstone Design, sponsored by Terumo Medical

Fall 2026

- Designing a multi-point temperature and pressure sensing probe that travels through a catheter heat-shrink lamination process, logs data, and reports through a custom GUI, for a team of BME, ECE and MEEG
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LEADERSHIP

Vice President of Judicial Affairs & Expansion | Interfraternity Council | *UD*

May 2025 - Present

- Managed compliance and policy execution for 29 chapters (1800+ members), utilizing strong organizational and persuasive communication skills to drive accountability and culture change.

2nd Counselor, Vice President | Sigma Pi Iota-Beta Chapter | *UD*

November 2024 - November 2025

- Directed internal relations by managing 21 chair positions and their respective committees, streamlining organizational processes to ensure effective execution of biweekly events.
- Assist in conducting weekly chapter meetings by outlining goals, facilitating discussions, and increasing morale.

SKILLS

Fusion 360, SOLIDWORKS, Python, Matlab, Image J, JMP Statistical Software, Soldering, Laser Cutting, Breadboard Circuits, Woodshop, Milling, CNC, Agentic Engineering, MCP Integrations, GD&T / tolerance stack-up (ASME Y14.5)

CERTIFICATIONS

MATLAB, Simulink, and Signal Processing Onramp; CITI Human Subjects, RCR, GCP

PERSONAL INTERESTS

Skiing, Golfing, Backpacking/Hiking, Mixing music, Weightlifting, International Travel